Geosteering Under Structural Uncertainty

Project: GeoSteering Structural Uncertainty
Synthetic replay run and simplified causal comparison. Training and portfolio tool; not validated for live operations.
Run ID: 4c02bea8-033c-4c7e-90c1-1586a971daa7
Status: PARTIAL / exploratory; revealed MD (ft): 2050.0
scenario_version: structural_dip_01-v1
seed: 123
code_revision: 9176b26a74972cdcaa8d0b41d1737bfa2c3fe8a3
dirty_working_tree: True
coordinate_reference: LOCAL_EAST_NORTH_FT; LOCAL_TVD positive down

Data summary and QC

Well	Curves	QC valid	Warnings
active_01	GR, RES	True	
offset_01	GR, RES	True	
offset_02	GR, RES	True	
offset_03	GR, RES	True	

Input provenance (SHA-256)

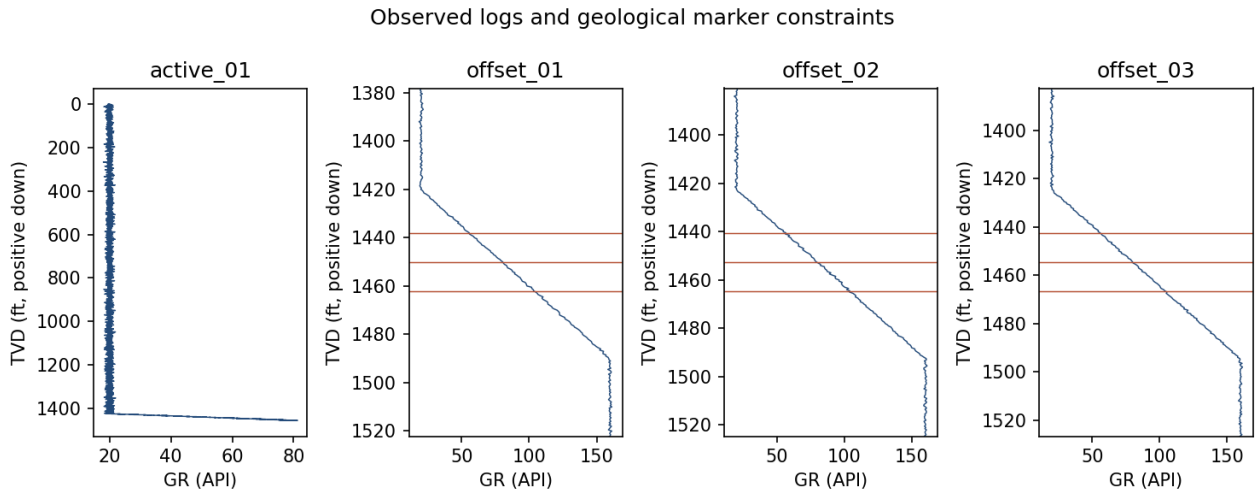
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Correlation and markers

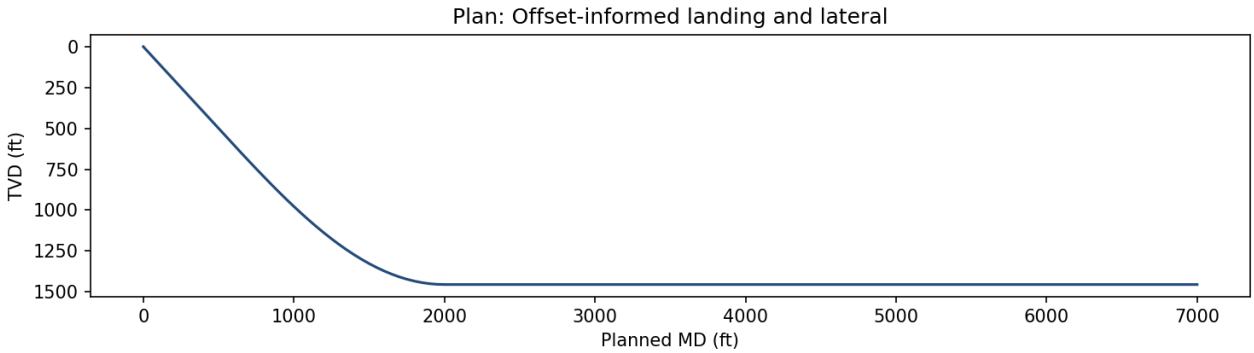
Offset formation picks constrain a regional estimate in shared along-track x and local TVD. Active observations and markers are limited to the saved revealed depth.

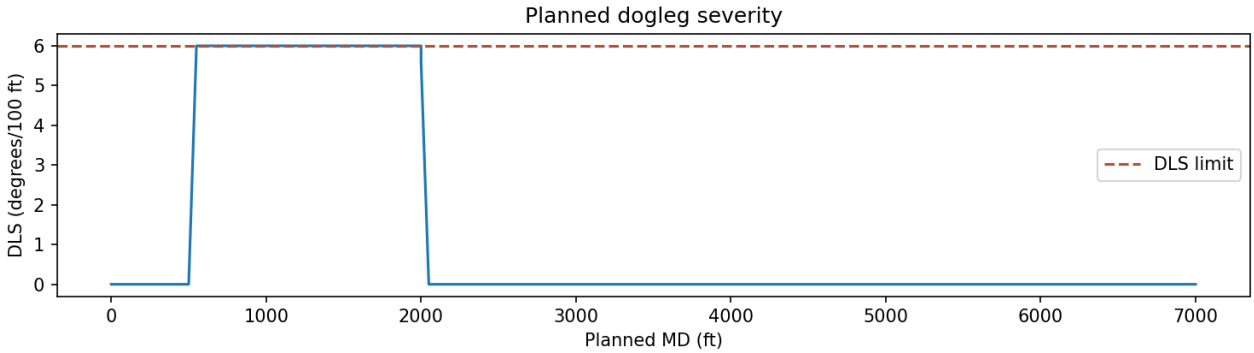
Well	Marker	MD (ft)
offset_01	Top_A	1438.16
offset_01	Mid_A	1450.16
offset_01	Base_A	1462.16
offset_02	Top_A	1440.66
offset_02	Mid_A	1452.66
offset_02	Base_A	1464.66
offset_03	Top_A	1442.66
offset_03	Mid_A	1454.66
offset_03	Base_A	1466.66



Pre-spud plan and DLS

Plan name: Offset-informed landing and lateral; Plan ID: 9e9ce661-cf60-41ce-bebb-1af2690060e1
Max DLS constraint: 6.0 degrees/100 ft; exceedance count: 0



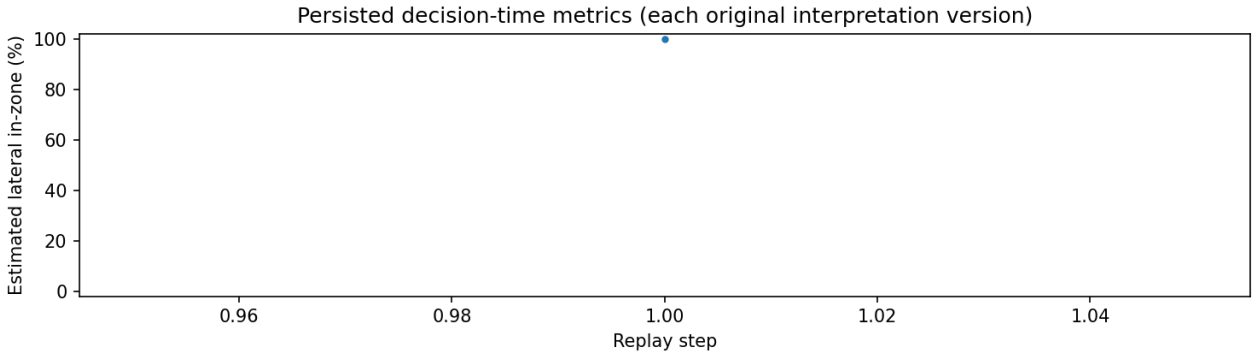


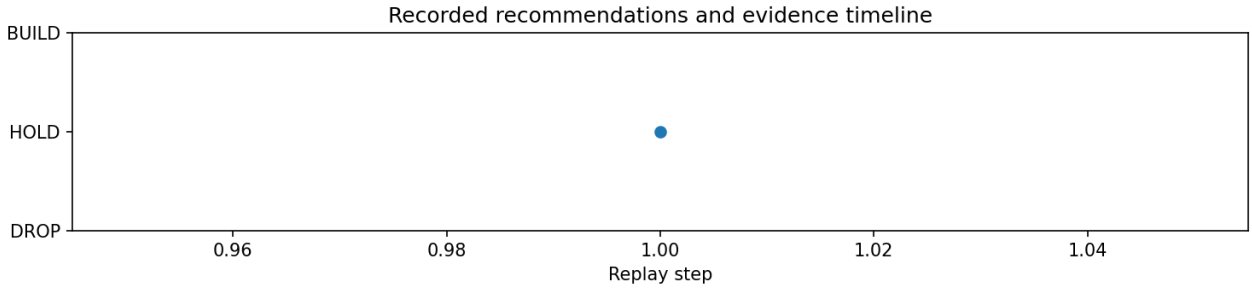
Steering decisions and KPI results

Replay recommendations do not change the recorded trajectory. Estimated metrics are frozen at each reveal using the cited interpretation, and exclude the build/landing interval.

Step	MD interval (ft)	Action	Evidence / interpretation version
1	2000-2050	HOLD	Acquired landing MD 2000 ft and synthetic GR near 80 API support HOLD.; version 6ad2afe2-31c0-464a-8a0f-15c2647ab480

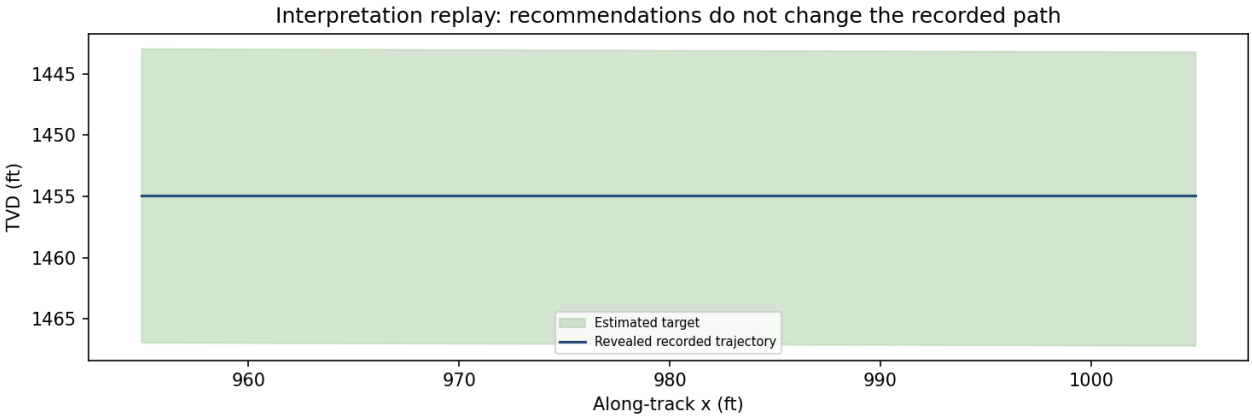
Step	Revealed MD (ft)	Estimated in-zone (%)	Mean center distance (ft)
1	2050	100.000	0.125





Final interpretation

Interpretation version: 6ad2afe2-31c0-464a-8a0f-15c2647ab480; rationale: Carry forward current interpretation: Acquired landing MD 2000 ft and synthetic GR near 80 API support HOLD.
Assumed uncertainty band (ft): 10.0. This scenario-assumed range is not a calibrated statistical confidence interval.



Independent evaluation

Withheld truth evaluation unavailable until completion. This partial report contains estimated metrics only.

Assumptions and limitations

Synthetic calibration: GR = 80 + 2*(TVD - target center) API, clipped to 20-160 API, with 0.6 API Gaussian noise. This deliberately signed layer signal is known to the synthetic operator; real gamma-ray inversion does not generally provide this information. Saturation cannot identify an exact center and is logged as unresolved evidence.
Scoring: estimated zones use exact piecewise-linear MD interval integration. Nonlinear withheld geology uses at most 1-ft intervals with boundary-crossing interpolation; reference refinement to 0.1 ft changes in-zone results by less than 0.001 percentage points. The final-point indicator is not an interval-length percentage.
Coordinates: feet; MD along the well, local TVD positive down, along-track x eastward. Minimum-curvature integration uses survey inclination and azimuth. Layer metrics are length weighted over the lateral interval with an explicit evaluated-length denominator.
The operator baseline uses offset picks only. Subsequent interpretations may use only revealed active measurements. Geological truth remains fixed and is isolated from live interpretations and recommendations.
Causal trials share geology, initial state, seed and trajectory constraints. HOLD/BUILD/DROP use a simplified bounded kinematic response. Report measured outcomes, including failure; no production drilling, sensor physics or operational safety claims are made.
Idealized 2D geology and synthetic GR/resistivity response; sparse offsets and assumed uncertainty. No real-time telemetry or WITSML support. Numerical summaries are reproducible; PDF timestamps and run identifiers can differ.